1. **Research Questions:**
    a. Given this data set on the historical prices of gold and other economic factors, to what extent can a multiple linear regression model utilizing the S&P 500 Index, crude oil prices for the United States, the EUR/USD exchange rate, and silver prices significantly predict the price of gold?
    b. Which of these exploratory variables is the most significant indicator of gold prices?
    c. Is there a certain time of year (fiscal quarters 1-4) that you would benefit the most from investing in gold?

All of the analysis will be performed with a confidence interval of 0.95 and an alpha = 0.05.

The answers to these questions could be valuable as investors and analysts seek to understand the drivers of gold prices to manage risk and portfolios more effectively.

2. **Describe The Data**

The data used in this analysis is from kaggle and can be found here: https://www.kaggle.com/datasets/sid321axn/gold-price-prediction-dataset

For my analysis I will be extracting 6 columns from this data set to answer the proposed questions above:

Column Name	Data Type	Description
Date (Date)	DateTime	Date of the recorded data (this will be transformed into a categorical variable representing each quarter (1-4) in a fiscal year. This is denoted as the extra column in the bottom of this table "Quarter"
Adj_Close (Gold)	Numerical	This is the adjusted closing price of gold on a given day
SP_Ajclose (SP500)	Numerical	This is the adjusted closing price of the S&P 500 on a given day
EU_Price (Euro)	Numerical	This is the exchange rate of USD : Euro on a given day
USO_Adj Close (Oil)	Numerical	This is the adjusted closing price of oil on a given day
SF_Price (Silver)	Numerical	This is the adjusted closing price of silver on a given day
Quarter	Categorical	A categorical variable generated from the Date column to represent each quarter in a physical year (1,2,3,4)

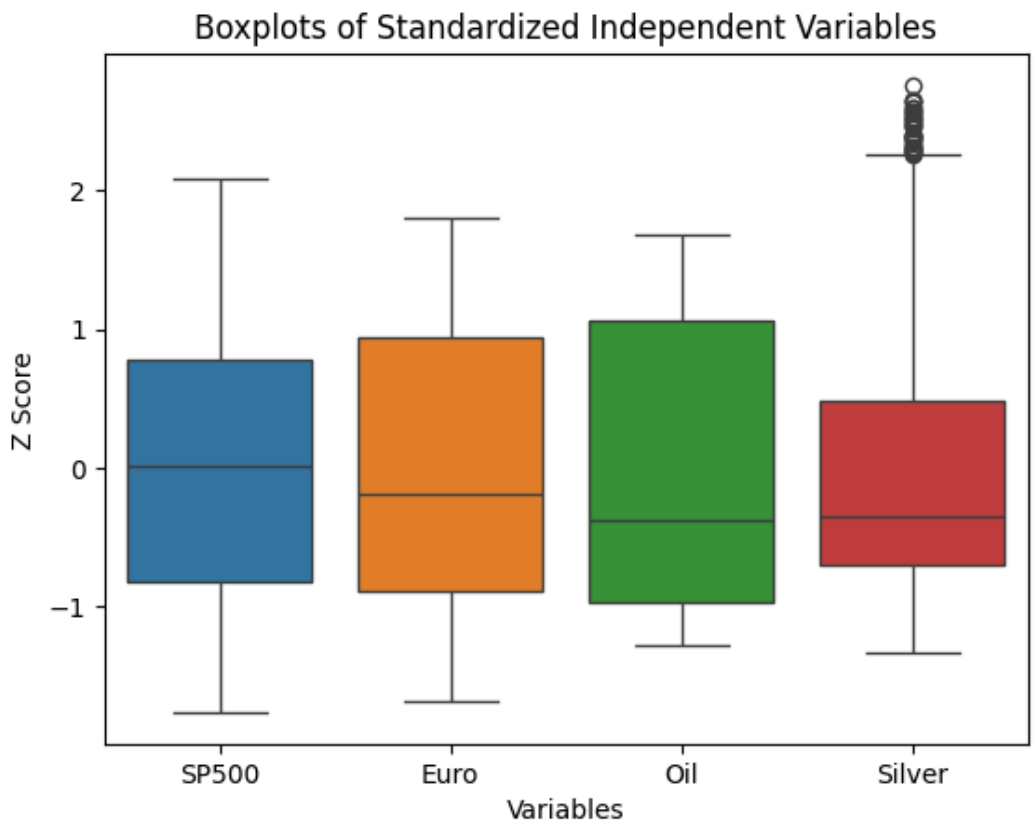
The adjusted closing price for these variables was chosen over the raw closing price because it is a more accurate representation of the value of the commodity or stock as it adjusts for corporate actions like dividends and stock splits. A simple random sample of 1000 was taken

from this data set to perform this analysis. To clean the data I renamed all the columns to those seen in parentheses in the above table under “Column Name”. All of the numerical values except for gold were standardized via z-score scaling to ensure the coefficients in the linear regression model are comparable. As described in the table the “Quarter” variable was generated by grouping each event by date and categorizing it by the time of year based on fiscal quarters. Any “NAs” were also removed from the data set during the cleaning process.

3. **Statistical Methods**

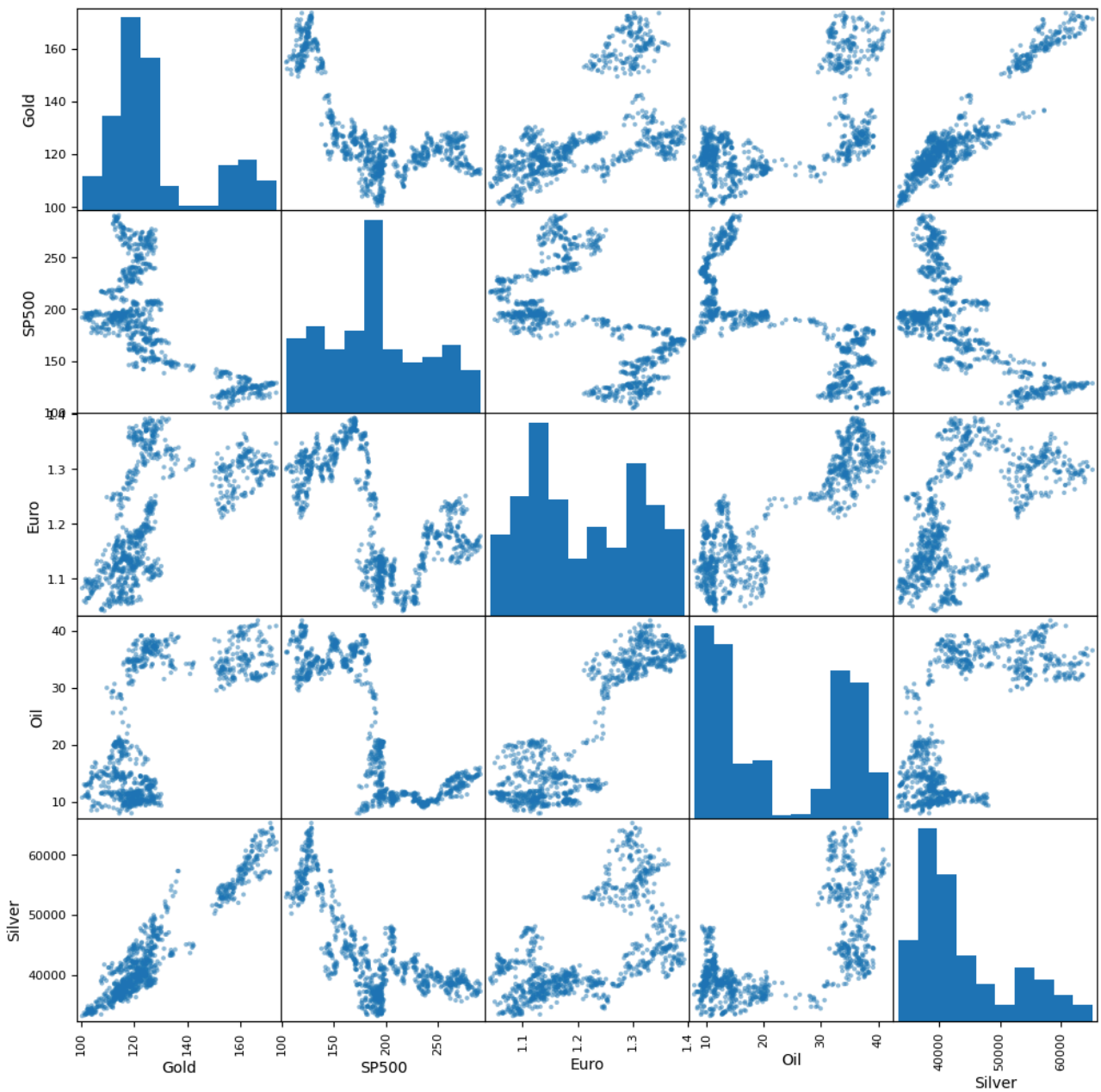
   I plan to use a simple multiple linear regression to fit the 4 numerical exploratory variables (SP500, Euro, Oil, Silver) against the response variable (Gold). I will then assess the significance of this fit. I also plan to perform a one way analysis of variance to see if there is a significant relationship between Gold price and time of the year (Quarter) as a categorical variable. To ensure the validity of these models, I will also perform diagnostic tests for linearity, normality of residuals, and independence of errors (Durbin-Watson).

4. **Results**



The figure above shows a boxplot of the standardized independent variables. This validates that the standardization was successful and enables the comparison of each

variable coefficient in the model to see which is the most impactful on the price of gold. It is also important to note that there does not appear to be any large outliers in any of the variables that could pull the model in one direction or the other. Silver has a couple high outliers around the +3 standard deviation range, this should not drastically affect the model but it is best practice to confirm this through statistical analysis which we will do later in this section.



The above figure shows a scatter matrix, presenting each relationship between the variables and a histogram of the spread of data in each variable. Silver and gold show an extremely

strong positive linear correlation while the rest, Oil, Euro and SP500 show more moderate linear trends. The internal scatter plots (representing correlations between independent variables) show no signs of extreme multicollinearity suggesting that each predictor operates independently which is a crucial assumption made when fitting a linear regression model. The histograms through the figure so the random sample captured a broad distribution of market values. Many of the variables show a bimodal distribution which is common in financial statistics as each peak represents a different market time.

Linear Regression Model Fit Results:

Adjusted R^2	F-Statistic	F-Critical	P-Value	Alpha
0.908	2464	2.38	<0.01	0.05

The table above shows a summary of the model. With an F-Statistic > F-Critical and P-Value < Alpha the null hypothesis ($\beta_1+\beta_2+\beta_3 = 0$) is rejected and the alternate hypothesis ($\beta_1+\beta_2+\beta_3$!= 0) is accepted. After determining the model is able to predict the price of gold with some level of significance, analysis of the impact of each variable on the model was performed and is described below.

Variable	Coefficient	P-Value	95% Confidence Interval
SP500	-0.8588	1.866352e-02	[-1.574, -0.144]
Euro	2.3280	1.252052e-07	[1.470, 3.186]
Oil	-2.5222	1.287079e-05	[-3.651, -1.393]
Silver	16.9759	0.000000e+00	[16.457,17.495]

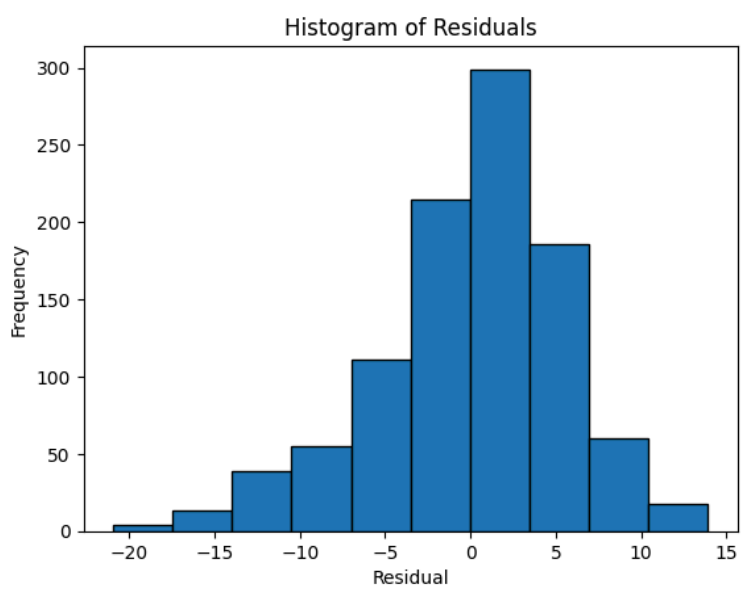
With each coefficient's P-Value < 0.05 (alpha), it can be stated that all of our variables are significant in this model. Looking at the coefficients, there are 2 that are negatively correlated to the price of gold (SP500 and Oil) and 2 variables that are positively correlated with the price of gold (Euro and Silver). It should be noted that the P-value and correlation coefficient of silver indicate an extremely close relationship to gold prices (as seen in the scatterplot above). It is possible that this relationship is too close and silver could be serving as a proxy to gold prices in this model. The concern here being that silver is so closely related to gold it is drowning out the impact of all the other coefficients in the model. Because of this concern it was determined that the model should be re-fit without silver. A comparison of the 2 models is shown below.

Fit Results with and without silver:

Metric	Model 1 (with silver)	Model 2 (without Silver)
Adjusted R^2	0.908	0.527
F-Statistic	2464	371.3
P-Value	0.00	8.04e-162

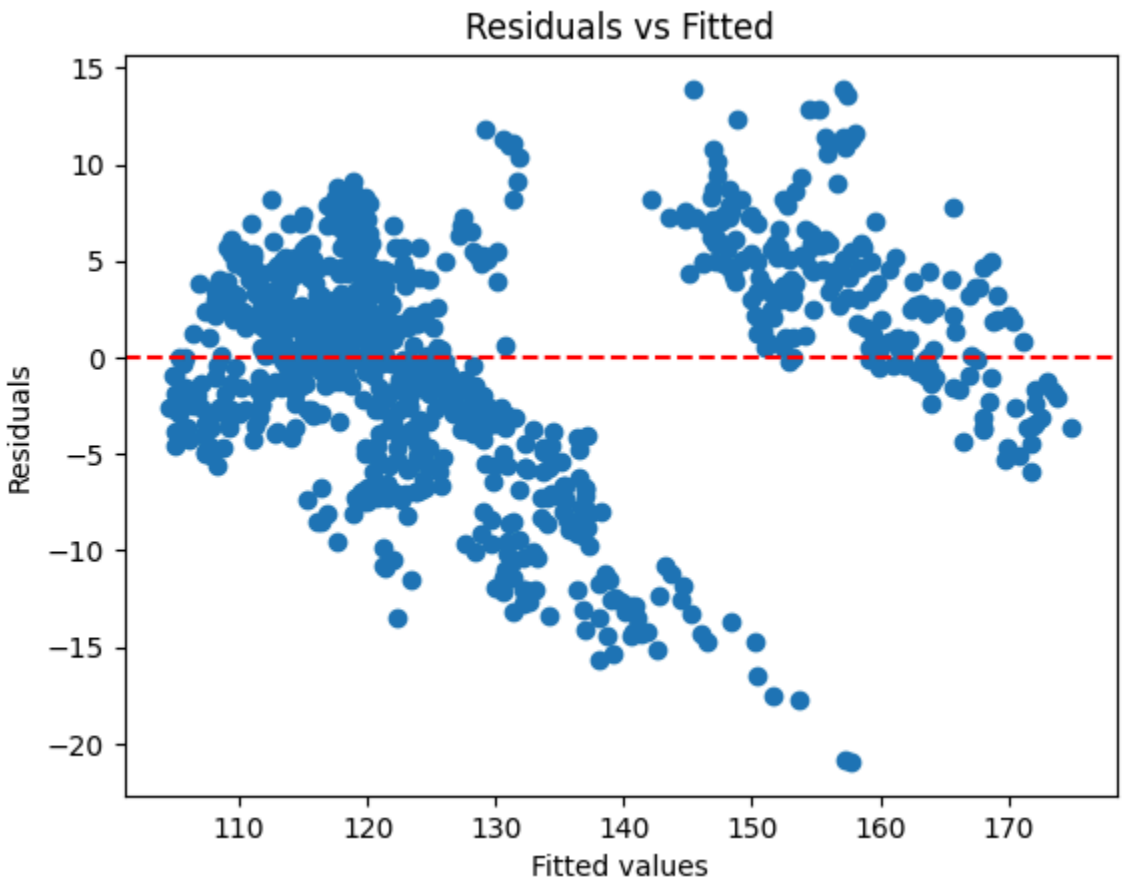
As you can see in the table above both models are able to predict the price of gold to some level of significance (compared to alpha= 0.05) but the large drop in Adjusted R^2 for the second model shows how much silver was supporting the first model. An analysis of the individual coefficients for the second model can be seen below. All three variables are significant predictors based on the P-Values when compared to alpha = 0.05. The large increase in coefficient between the first and second model for the variables confirms the suspicions that silver was drowning out the relationships of the other variables to the price of gold. Lastly, the Durban Watson statistic for this model is 2.007 which strongly suggests each event's residual is independent of itself (there is no autocorrelation). The "Histogram of Residuals" plot below also confirms that the normality of residuals assumption required for F test and t tests for significance is satisfied.

Variable	Coefficient	P-Value	95% Confidence Interval
SP500	-11.6045	2.323245e-50	[-13.05 , -10.16]
Euro	8.8658	2.139557e-19	[6.97 , 10.76]
Oil	-5.4630	2.892815e-05	[-8.02 , -2.91]

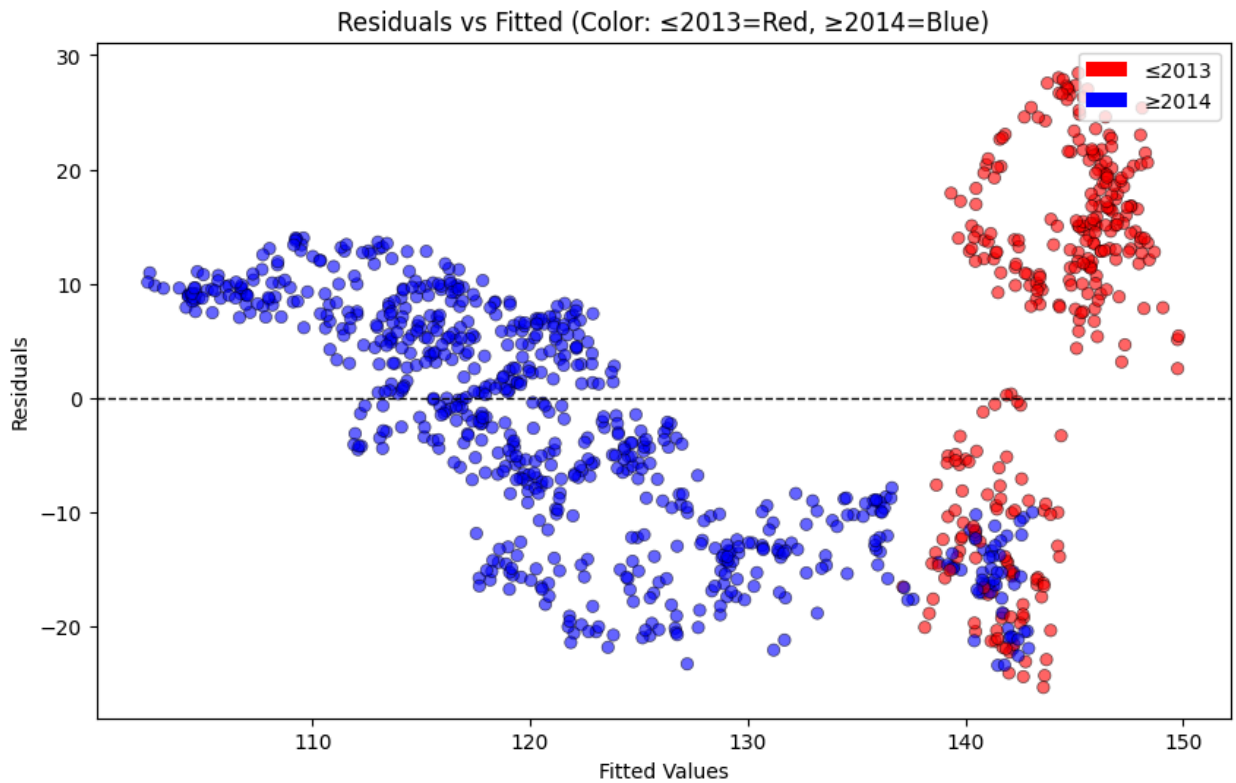


The data below dives further into the analysis of the residuals from the second multiple linear regression model to better understand its robustness and limitations:

Below is a plot of the residuals. Two clear clusters, both with a downward trend can be seen in the plot. This is not the “random cloud” about the zero line you would expect to see with a robust model. These 2 different “populations” pointed to something fundamentally different about 2 parts of the data set. Further analysis revealed the cause behind this.

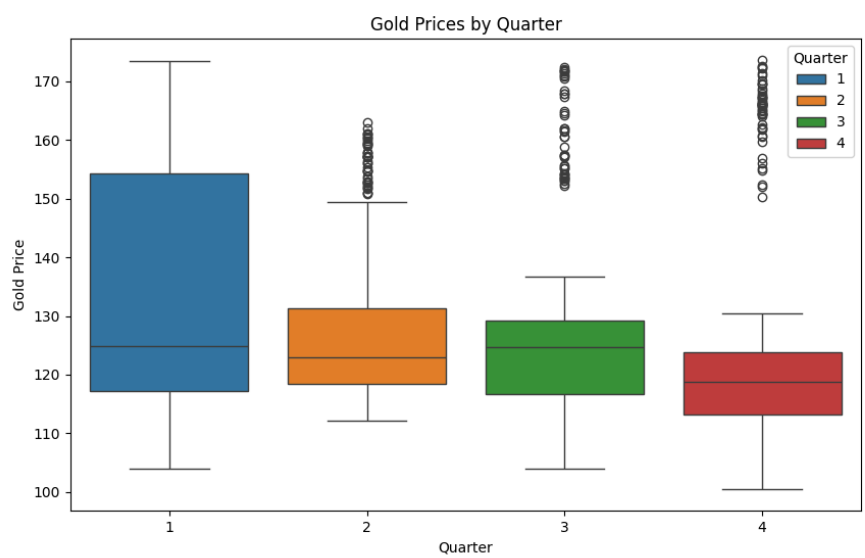


The data from the random sample taken to build this model consisted of observations between 2011-2018. Some research into major market shifts in gold during this time period revealed a historic crash in gold prices during the year 2013. Plotting the same residual plot as above but color coding it by 2013 and earlier (red) vs 2014 and later (blue) reveals 2 distinct populations of residuals, those before and those after the gold crash in 2013. The red dots in the top right of the plot represent an under evaluation of the price of gold during the years 2013 and earlier. It would make sense that the model under-predicts the price of gold during that time as the model is weighted toward the post crash valuation because there are more observations after 2013 than before. There was a “premium” priced into gold pre-crash that the model does not account for, this premium went away after 2013 and thus you see the second population in blue.

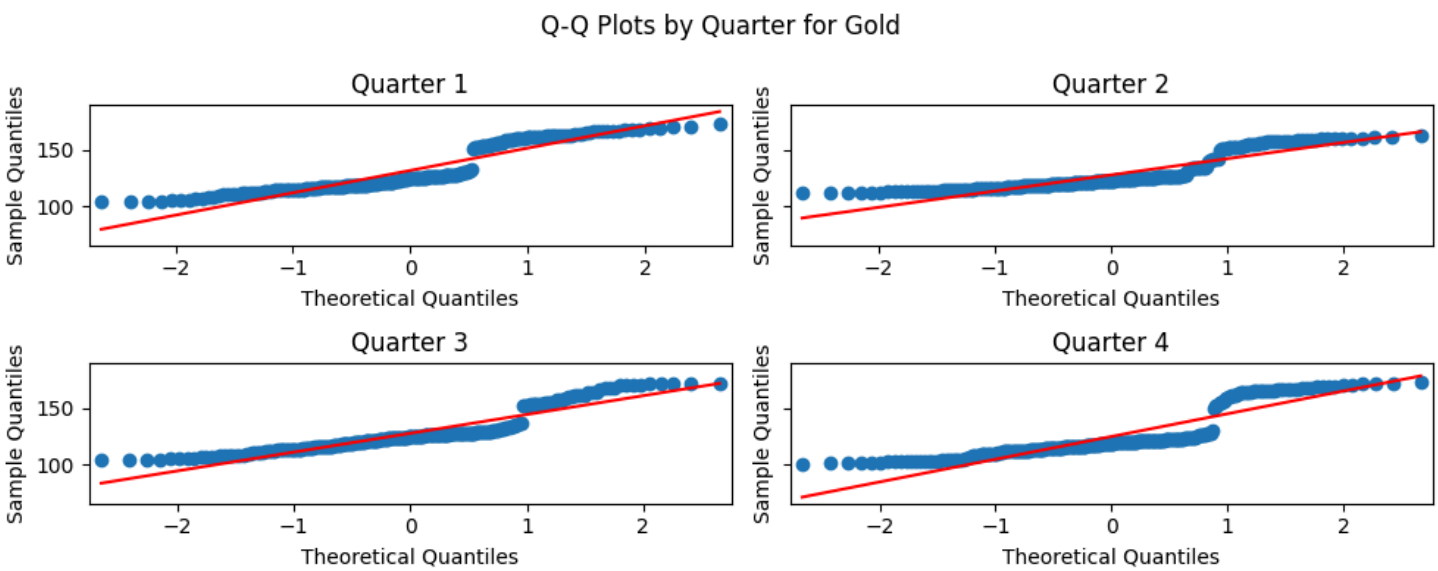


One-Way ANOVA Quarterly analysis:
Prior to running the ANOVA for gold prices as a function of fiscal quarter, a boxplot and Q-Q plot were generated to understand if the data was fit for this kind of analysis.

Quarter	1	2	3	4
Standard Deviation	19.96	14.48	16.79	20.29



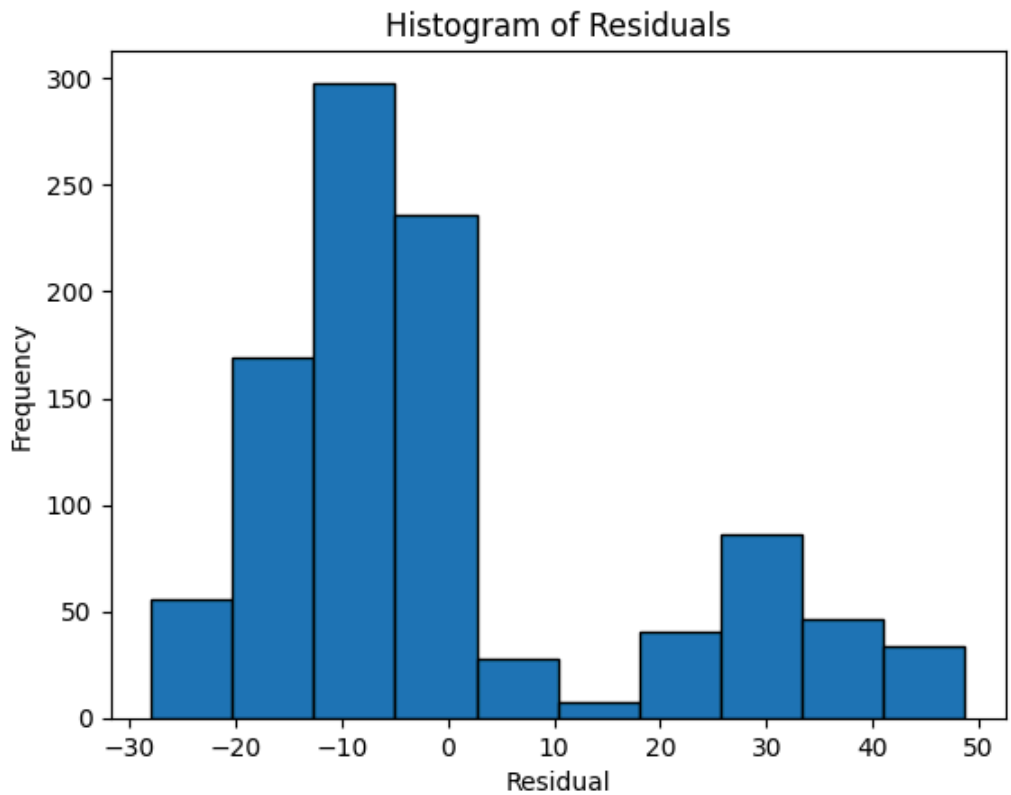
The boxplox and standard deviation table above provide confidence for homogeneity of variance in the 4 groups of data (no SD is 2x another) which is necessary when performing ANOVA.
The Q-Q plots for each quarter below show that each population is fairly normally distributed about the mean, enough so that we can be confident in the results of this ANOVA test.

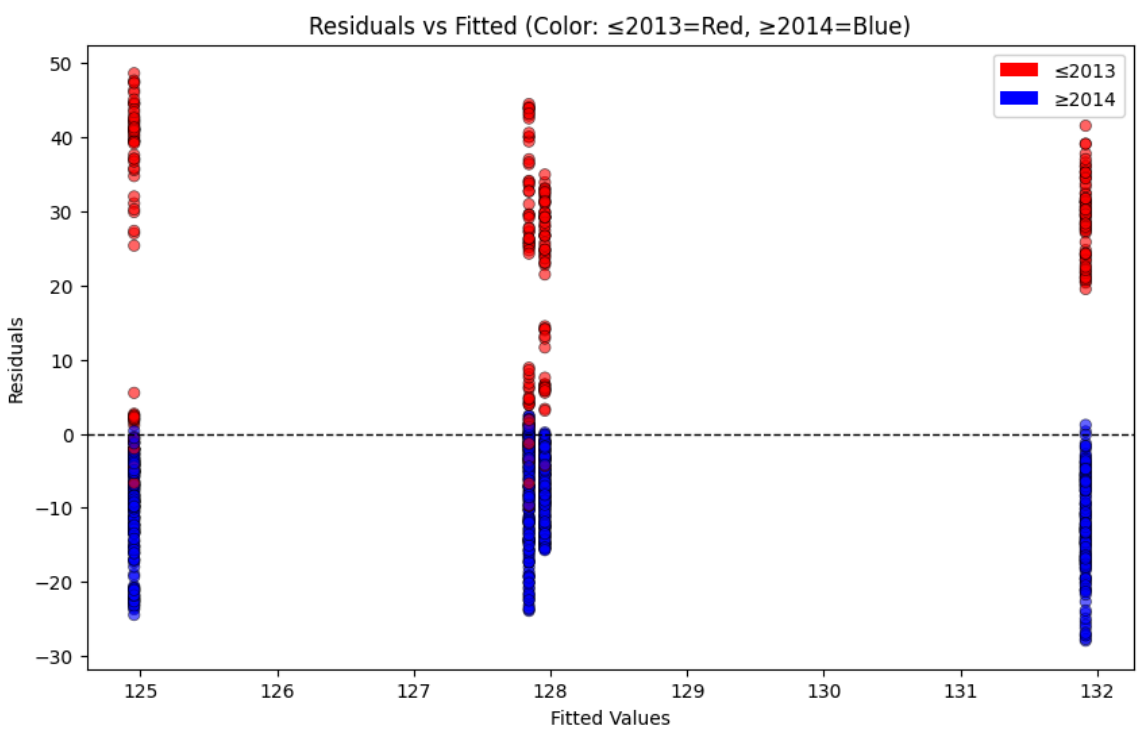


ANOVA Results:
The ANOVA results partitioned the total variation in gold prices into two components: the variation between quarters (SS = 6,016.98) and the residual variation within quarters (SS = 323,357.16). While the unexplained variation is much larger, the portion of variation explained by the change in quarters was sufficient enough to produce a significant F-statistic (6.177). Comparing this to our F-critical value of 2.61 confirms that the quarterly means are spread further apart than would be expected if time of year had no impact on price. This is further supported by the reported P-Value of 0.000369 which is less than alpha = 0.05. After this finding a pairwise Tukey's test was performed to understand the relationship between each quarter. The results are in the table below:

group1	group2	meandiff	p-adj	95% CI	reject H_0
1	2	-3.9568	0.0729	[-8.15 , 0.24]	False
1	3	-4.0731	0.062	[-8.28 , 0.14]	False
1	4	-6.9623	0.0001	[-11.14 , -2.79]	True
2	3	-0.1163	0.9999	[-4.24 , 4.01]	False
2	4	-3.0055	0.2326	[-7.10 , 1.09]	False
3	4	-2.8892	0.2682	[-6.99 , 1.21]	False

Further Analysis of this model however shows the same findings as the multiple linear regression model. The residual plot and histogram below clearly show the 2 populations of data (before and after the 2013 gold crash).





The effect of the 2013 gold crash is actually more pronounced in this model because of the nature of a one-way ANOVA. With only one exploratory variable (time of year) the model leaves the crash entirely unexplained. This is in contrast to the multiple linear regression model where other macro economic factors accounted for **some** of the price variance during that time. Again, we see large positive residuals for the red dots indicating an under prediction by the model for gold prior to the crash.

5. Conclusions and Limitations

Multiple Linear Regression Model:

After comparing both models (with and without silver) it was determined that the second model would be more useful in the context of predicting gold based on other macroeconomic factors. This is because there is not much gained by keeping silver in the model (it is intuitive to assume that the price of gold and silver would mirror each other as they are both precious metals). By removing silver we are able to magnify the "responsibility" the other variables have for predicting the price of gold which is more useful in this specific case. In conclusion it can be said that with 95% confidence that for every 1 standard deviation the S&P 500 increases, gold drops between $13.05 - $10.16. Every time the Euro:USD ratio changes by 1 standard deviation in favor of the Euro the price of gold increases between $6.97 - $10.76 and every time the price of oil increases by 1 standard deviation the price of gold drops between $8.02 - $2.91. The large coefficient of -11.6045 and low P-Value of 2.323245e-50 for the S&P 500 makes it the most significant predictor of gold prices out of the 3 in the second model.

One-Way ANOVA:

While the ANOVA test confirmed there is some significance to the time of year and the price of gold, the Tukey's test proved that there is only a significant difference between Q1 and Q4. It can be said with 95% confidence that on average the price of gold is between $2.79 - $11.14 lower in Q4 than it is in Q1. This finding could be a valuable insight for investors who are looking to move money into or out of the gold market during a certain time of year.

It should be noted that both of these models relied on a simple random sample of 1000 days. While a sample size of 1000 is statistically sufficient, the result may vary if different days were sampled. Things like geopolitical instability, or other major historical economic events (like the 2013 gold crash) can drastically shift the predictability of this model. Another limitation is the omitted variables. There are many driving factors in the price of gold and only a few were examined in this exercise. Other factors like Federal Reserve interest rate changes or global inflation rates are not accounted for.

With the goal of the multiple linear regression model to forecast current/near-future prices in gold it would likely benefit from more recent data, preferably 2021-present (to exclude the market crash during covid when gold saw a large surge in prices). I would expect this to resolve

much of the bimodal residual distribution seen in both of these models. Gold's relationship between the Euro, Oil and the S&P 500 is constantly changing so capturing this in one linear model is probably not the best fit.